

Reviewer Pack — Minimal Number of Invariant Points and Communication Complex...

Entry e132a7fc4394 · INCONCLUSIVE

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Minimal Number of Invariant Points and Communication Complexity

Entry ID: e132a7fc4394 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 08:40:07 UTC

1. Conjecture

Field A (mathematical branch): Invariant Theory **Field B** (complexity object): Communication Complexity

Statement:

For any given boolean function $f(x_1, x_2, \dots, x_n)$ with n inputs, the minimal number of invariant points under the action of the symmetric group S_n on the set of all inputs (denoted as $\min_invar(f)$) is linearly correlated with its deterministic communication complexity $c(f)$, such that $\min_invar(f) = \Theta(c(f))$.

Rationale (proposer's reasoning):

Invariant Theory studies the invariants of geometric objects, and the number of invariant points can be used to characterize the structure of these objects. Communication Complexity measures the amount of communication needed to compute a boolean function, which is related to the complexity of the function itself. By linking these two concepts, we may uncover new insights into the computational hardness of functions.

Taxonomy category: Invariant Theory × Communication Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1ea97898e1533914

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given boolean function f with n inputs, the conjecture is supported if and only if the correlation coefficient between $\min_invar(f)$ and $c(f)$ exceeds 0.9 for at least 80% of the generated functions, where each function has a p-value less than 0.05 from a statistical test.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Invariant Theory" AND "Communication Complexity" AND boolean function" - "min_invar(f)" AND symmetric group S_n AND communication complexity" - deterministic communication complexity AND linear correlation WITH invariant points

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def symmetric_group_actions(n):
        actions = []
        for i in range(n):
            action = list(range(n))
            action[i:] = action[i+1:] + action[i:i+1]
            actions.append(action)
        return actions

    def communication_complexity(f, n):
        # Placeholder function to simulate communication complexity
        # This is a dummy implementation and should be replaced with actual algorithm
        return len(f) # Simplified for demonstration purposes

    def min_invariant_points(actions, f):
        n = int(math.log2(len(f)))
        invariant_points = []
        for x in range(2**n):
            if all(f[x] == f[action[x]] for action in actions):
                invariant_points.append(x)
        return len(invariant_points)
```

```

def correlation_coefficient(values1, values2):
    n = len(values1)
    mean1 = sum(values1) / n
    mean2 = sum(values2) / n
    cov = sum((values1[i] - mean1) * (values2[i] - mean2) for i in range(n)) / n
    var1 = sum((values1[i] - mean1)**2 for i in range(n)) / n
    var2 = sum((values2[i] - mean2)**2 for i in range(n)) / n
    return cov / (math.sqrt(var1) * math.sqrt(var2))

results = []
for _ in range(30):
    n = random.choice([5, 10, 15, 20, 30, 40])
    f = generate_boolean_function(n)
    actions = symmetric_group_actions(n)
    c_f = communication_complexity(f, n)
    min_invar_f = min_invariant_points(actions, f)
    results.append((c_f, min_invar_f))

if not results:
    return {
        "metric_name": "correlation_coefficient",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "empty_results"
    }

c_f_values, min_invar_f_values = zip(*results)
corr_coeff = correlation_coefficient(c_f_values, min_invar_f_values)

return {
    "metric_name": "correlation_coefficient",
    "metric_value": corr_coeff,
    "instances_tested": len(results),
    "n_max": max(len(f) for _, f in results),
    "conjecture_holds": corr_coeff > 0.9,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}}, **result}}")
        results.append(result)

    mean_corr_coeff = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_corr_coeff} std=0.0 support_fraction=1.0")

```

```

elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_corr_coeff} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next((seed for seed, r in zip(seeds, results) if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=\"correlation_coefficient\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b7adfc4.py", line 91, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b7adfc4.py", line 60, in run_trial
    min_invar_f = min_invariant_points(actions, f)
                  ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b7adfc4.py", line 41, in min_invariant_points
    if all(f[x] == f[action[x]] for action in actions):
        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b7adfc4.py", line 41, in <genexpr>
    if all(f[x] == f[action[x]] for action in actions):
        ~~~~~^
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete its execution to provide a correlation coefficient and p-value as required by the support condition.
 | next: Re-run the test with proper error handling to ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15262
2	propose	ollama_remote	glm4:latest	0	0	12767
3	preregistration	ollama_remote	glm4:latest	0	0	17934
4	novelty	ollama_remote	glm4:latest	0	0	27172

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
5	novelty	ollama_remote	glm4:latest	0	0	23073
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23339
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19372
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15565
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21148
10	judge	ollama_remote	glm4:latest	0	0	12374

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 188006 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/e132a7fc4394.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e132a7fc4394.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/e132a7fc4394.tar.gz` (if generated)